

Exploratory Data Analysis

This dataset contains browsing sessions from an online shopping website. Each row represents one session, and our goal is to predict whether the session ends in a purchase (**Revenue = True**) or not (**Revenue = False**) based on the visitor's behavior.

Before training any models, we explored the dataset to understand its distributions, compare purchasing and non-purchasing sessions, and identify relationships between features.

Target Distribution

The target variable is imbalanced:

Revenue	Percentage
False (No Purchase)	84.53%
True (Purchase)	15.47%

Most sessions do not end in a purchase.

This is important for the modeling stage because accuracy by itself can be misleading. For example, a model that predicts **No Purchase** for almost every session could still achieve around 84% accuracy without actually learning how to identify purchasing sessions.

For this reason, we will pay attention to metrics such as Precision, Recall, and F1-score when evaluating the models.

Feature Distributions

ProductRelated

ProductRelated represents the number of product-related pages viewed during a session.

Statistic	Value
Mean	31.73
Median	18
Maximum	705

The mean is noticeably higher than the median, which indicates a right-skewed distribution.

Most sessions involve a relatively small number of product pages, while a smaller number of sessions contain much heavier browsing. These larger values create the long right tail and pull the mean upward.

This gives us an initial idea about how visitors normally browse the website.

PageValues

PageValues represents the value associated with the pages viewed during a session in relation to conversion.

At least 75% of the sessions have a **PageValues** value of 0.

So, most sessions have no recorded page value, while a smaller group has non-zero values. This becomes more interesting when we compare PageValues between sessions that resulted in a purchase and those that did not.

Feature vs. Purchase Behavior

After looking at the individual feature distributions, we compared the features between purchasing and non-purchasing sessions.

The purpose here was to understand whether there are noticeable differences in visitor behavior depending on whether a purchase happened.

ProductRelated and ProductRelated_Duration

Feature	No Purchase	Purchase
ProductRelated	29.07	48.22
ProductRelated_Duration	1076.26	1877.52

Purchasing sessions viewed more product-related pages on average and also spent more time on those pages.

The two results support the same pattern. Visitors who eventually purchased generally interacted more deeply with the product section of the website.

This is what we mean by **engagement** in this context. We are looking at how much the visitor interacts with the website through things such as pages viewed and time spent browsing.

PageValues

Revenue	Average PageValues
No Purchase	2.03
Purchase	27.65

There is a large difference between the two groups.

Although most sessions have a PageValues of 0, purchasing sessions have a much higher average value.

This suggests that higher PageValues are strongly associated with purchase behavior, making this feature potentially useful for our prediction task.

It is important not to confuse association with causation. The data shows that the two variables are related, but this analysis does not prove that a higher PageValues value causes a visitor to purchase.

BounceRates and ExitRates

Feature	No Purchase	Purchase
BounceRates	0.02317	0.00508
ExitRates	0.04549	0.01957

Both BounceRates and ExitRates are lower for purchasing sessions.

In general, non-purchasing sessions show more signs of visitors leaving the website or not continuing their browsing.

This fits with what we saw in the previous features. Purchasing sessions tend to involve more product browsing, more time spent on the website, and lower rates of early exit.

Looking at these features together gives us a clearer picture than looking at any single graph by itself.

Visitor Type

For categorical features, we used the purchase rate within each category rather than calculating averages from the encoded values.

Visitor Type	Purchase Rate
New Visitor	24.93%
Other	19.75%
Returning Visitor	14.09%

New visitors have the highest purchase rate in this dataset, while returning visitors have the lowest.

This is somewhat unexpected because we might assume that returning visitors would be more likely to purchase. However, the analysis is based on the actual observations in this dataset rather than that assumption.

Weekend

We also compared the purchase rate between weekday and weekend sessions.

Session Type	Purchase Rate
Weekday	15.08%
Weekend	17.45%

Weekend sessions have a slightly higher purchase rate.

However, the difference is relatively small, so we would not consider **Weekend** by itself to be a strong indicator of purchase behavior.

Correlation Analysis

We also examined the correlation between numerical features.

Correlation describes the strength and direction of a linear relationship between two variables. Its value ranges from -1 to +1.

The strongest relationships we found were:

Feature Pair	Correlation
ProductRelated / ProductRelated_Duration	0.880
BounceRates / ExitRates	0.901
Informational / Informational_Duration	0.632
Administrative / Administrative_Duration	0.608

The high correlation between **ProductRelated** and **ProductRelated_Duration** makes sense. Sessions with more product pages viewed generally involve more time spent on those pages.

The same applies to **BounceRates** and **ExitRates**. Sessions with higher BounceRates tend to also have higher ExitRates.

These high correlations tell us that some features contain overlapping information. This is something we need to keep in mind during preprocessing and feature selection, especially for models such as Logistic Regression where correlated inputs can lead to multicollinearity.

We are not removing features simply because their correlation is high. We will consider this later when deciding how to prepare the data for each model.

Overall EDA Story

After comparing the different features, we found a fairly consistent pattern in the dataset.

Purchasing sessions generally show stronger engagement with the website. They tend to involve more product pages, more time spent on those pages, higher PageValues, and lower BounceRates and ExitRates.

We also found that New Visitors had a higher purchase rate than Returning Visitors in this dataset, while the difference between weekday and weekend sessions was relatively small.

At the same time, the target is heavily imbalanced, and some features are strongly correlated with each other.

These findings will be useful when we move to the next stages of the project, particularly preprocessing, feature selection, model training, and model evaluation.

Main Findings

1. The target is highly imbalanced, with only 15.47% of sessions resulting in a purchase.
2. Purchasing sessions tend to involve more product pages and more time spent browsing them.
3. PageValues is substantially higher in purchasing sessions and shows a strong association with purchase behavior.
4. Non-purchasing sessions have higher BounceRates and ExitRates.
5. New Visitors have the highest purchase rate among the VisitorType categories.
6. Weekend sessions have only a slightly higher purchase rate than weekday sessions.
7. Several features have strong correlations, particularly **ProductRelated** with **ProductRelated_Duration** and **BounceRates** with **ExitRates**.
8. These observations give us a better understanding of the dataset before moving into the ML stage.